

Project Action Plan

SmartAuto ERP – Automobile Spare Parts Distribution & Sales Management System

GROUP IS 24

BSc Information Systems

GROUP MEMBERS

Index Number	Name of the Student	Assigned Primary Role
24021059	K. I. U. Thisera (Induwara)	Sales Representative / Mobile POS Lead
24021067	R. A. S. Thivanka (Sashik)	Store Manager / Inventory & Procurement Lead
24020451	L. A. C. R. Jayamali (Chathumi)	Business Owner / Admin & BI Reports Lead
24020771	V. Pavalaraj (Varshika)	Customer Portal & Public Services Lead

EXECUTIVE SUMMARY

During the proposal defense evaluation, the panel raised valid and critical concerns regarding the excessive scope, lack of domain specialization, and feasibility of developing a generic, multi-industry ERP across four undergraduate members within one academic year. In response, Group IS 24 has executed a complete strategic pivot: transitioning from an unmanageable multi-domain platform into **SmartAuto ERP**—a purpose-built Automobile Spare Parts Distribution and Sales Management System. Non-essential modules (such as comprehensive HR, payroll, and open-ended multi-industry handlers) have been removed. The system is strictly refocused on real-world auto parts workflows: inventory management, vehicle-to-part compatibility lookup, on-field mobile POS for sales representatives, and customer self-service ordering.

PANEL FEEDBACK SYNTHESIS & EVALUATION ANALYSIS

Panel Critique	Detailed Evaluation Context	Risk Identified by Panel
1. Overly Broad Scope & Feasibility	The panel noted that attempting a generic ERP across 4 unrelated domains (pharmacy, grocery, hardware, retail) with 10+ modules is an unfeasible "Sisyphean task" for a 4-member undergraduate team.	Critical Risk Surface-level modules with negligible depth; guaranteed failure during code defense.
2. Reinventing the Wheel	Freely available, enterprise-grade open-source ERPs (e.g., ERPNext, Odoo) already exist for small businesses. Rebuilding a standard ERP without domain novelty lacks academic value.	High Risk Inability to defend system uniqueness; potential suspicion of reusing existing solutions.
3. Vague Domain & Business Sizing	"Small and Medium Enterprises" was ill-defined without clear parameters regarding catalog complexity, employee count, or business operational boundaries.	Scope Ambiguity Inability to establish realistic test cases or measurable user boundaries.
4. Superficial HR & Module Gaps	The panel observed that if an ERP claims to offer an HR module, tracking mere attendance is insufficient without complete payroll, performance reviews, and recruitment.	Design Flaw Mislabeling basic operational tasks as full-scale enterprise modules.
5. Missing Client & Vague Needs	Without a real customer context, features were being conceptualized theoretically rather than solving an authentic, ground-level business bottleneck.	Practicality Gap Developing disconnected features that fail to meet industry-specific realities.

COMPREHENSIVE ACTION TAKEN: THE STRATEGIC PIVOT

Following panel instructions, the group consulted supervisor **Mr. Tharindu Ranathunga** and restructured the project from the ground up. Below is the direct comparison of the changes adopted:

Previous Proposal (De-scoped)

De-scoped

- **Domain:** Generic, multi-industry ERP trying to cater to groceries, pharmacies, hardware, and retail outlets.
- **Scope:** 10+ broad modules (General HR, Payroll, Finance, SCM, Multi-tier Warehousing, Procurement).
- **Target:** Ambiguous SME sector without specific employee or transaction limits.
- **Core Problem:** Generic CRUD forms with superficial analytical capability and no domain edge.

SmartAuto ERP (Approved Pivot)

Approved Scope

- **Domain:** Automobile Spare Parts Distribution and Sales Management System exclusively.
- **Scope:** 4 practical operational modules: Mobile POS, Spare Part Inventory, Administration/BI, and Customer Portal.
- **Target:** Mid-scale auto spare parts distributors, field sales representatives, auto repair workshops (B2B), and walk-in buyers (B2C).
- **Core Novelty:** Vehicle Compatibility Finder (Brand > Model > Year > Engine) and on-field Mobile POS.

VERTICAL WORKLOAD DISTRIBUTION ACROSS GROUP MEMBERS

The system is distributed vertically so that each member takes full end-to-end ownership of their assigned user role, user interface screens, business logic, and database entities:

Member 1: K. I. U. Thisera

24021059

Primary Role: Sales Representative

- **Assigned Screens:** Sales Rep Dashboard, Mobile POS Terminal, Order Management, Customer Management, Invoicing, Sales History.
- **Core Responsibilities:** Field sales processing via mobile POS, retail & walk-in sales checkout, customer registration, invoice generation, payment recording.
- **Entities:** customers, sales, sale_items, orders, order_items, payments.

Member 2: R. A. S. Thivanka

24021067

Primary Role: Store Manager

- **Assigned Screens:** Product Management, Inventory & Stock Movements, Purchase Orders, Supplier Management, Spare Part Finder Logic.
- **Core Responsibilities:** Spare parts cataloging, stock in/out adjustments, low-stock threshold detection, vehicle compatibility matrix, supplier procurement.
- **Entities:** products, categories, brands, vehicles, compatibility, inventory, suppliers, purchase_orders.

Member 3: L. A. C. R. Jayamali

24020451

Primary Role: Business Owner / Administrator

- **Assigned Screens:** Authentication & RBAC, Owner Dashboard, Employee Accounts, Business Reports, Activity Logs, System Settings.
- **Core Responsibilities:** User authentication & role security, employee sales tracking, profit/sales/stock analytics, audit trails, system configuration.
- **Entities:** users, roles, employees, notifications, activity_logs, settings.

Member 4: V. Pavalaraj

24020771

Primary Role: Shop Customer / Public User

- **Assigned Screens:** Public Product Catalog, Vehicle-Based Spare Part Finder (Customer UI), Online Ordering Portal, Order Tracking, Delivery Status.
- **Core Responsibilities:** Customer part search by vehicle make/model, self-service order placement for repair shops, order delivery tracking.
- **Entities:** product_catalog, customer_orders, deliveries, compatibility (view).

EXPLICIT OUT-OF-SCOPE BOUNDARIES

To ensure strict feasibility within the undergraduate timeframe, the following features are confirmed as out of scope: (1) Integrated online credit card payment gateways; (2) Live GPS route tracking for sales reps; (3) Complex recruitment and payroll HR; (4) Offline sync; and (5) Multi-warehouse distribution.